Length-Dependent Quantization Fragility in Audio State Space Models

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Abstract

Selective state space models (SSMs) such as Mamba compress an entire sequence into a fixed-size recurrent state, updated at every step, while attention gives every token direct, uncompressed access to every other token. We ask whether this compression makes SSM-based audio encoders more fragile to post-training quantization than Transformer-based ones, and if so, which internal tensor is responsible. Using two official, task-matched checkpoints on Speech Commands V2 (Audio Mamba, AuM, and the Audio Spectrogram Transformer, AST), we find that standard Linear-layer quantization does *not* explain AuM's fragility: AuM's accuracy already collapses to near-chance from full precision alone, immediately past its $\sim$1s training length, leaving quantization nothing additional to degrade by top-1 accuracy. AST degrades far more gradually with length and shows a real, bit-width-specific W4A16 accuracy drop at 40–80s. We then isolate AuM's own internal SSM tensors (Δ, A, B/C, and the recurrent state h) one at a time at native length, where AuM still has accuracy headroom. A nonparametric bootstrap over 140 class-balanced clips shows the recurrent state is fragile from the mildest bit-width tested (83.6% at 8-bit, falling to 7.9% at 4-bit), Δ holds up through 6-bit (88.6%) before collapsing between 6- and 4-bit (19.3%), while A and B/C remain within noise of full precision at every bit-width tested (95–97%). Per-timestep state-divergence traces show these two failure shapes are qualitatively different: the state's error accumulates smoothly every step, while Δ's is heavy-tailed, with most steps barely perturbed and a minority blowing up. Code, data manifests, and all raw results are public.[1]

1 Introduction

Selective SSMs [Gu and Dao, 2023] process a sequence with linear time and memory complexity by compressing all past context into a fixed-size recurrent state h_t, updated at every step as $h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$ with $\bar{A}_t = \exp(\Delta_t A)$. This is structurally different from self-attention, which keeps every token's representation and lets any token attend to any other directly, with no compression bottleneck. Audio Mamba (AuM) [Erol et al., 2024] applies this architecture to spectrogram classification as a direct, bidirectional analogue of the Audio Spectrogram Transformer (AST) [Gong et al., 2021], making the two models an unusually clean pair for isolating the effect of the SSM's compressed-state design from everything else (patch embedding, classification head, training data).

Post-training quantization (PTQ) adds a second source of approximation error on top of a model's own architecture. We ask two questions. **H1:** does PTQ hurt AuM more than AST, and does that gap widen as input length grows past AuM's $\sim$1-second training length, where the state has to summarize far more than it was trained to? **H2:** if the SSM's internals are a distinct source of fragility, which specific tensor in the recurrence – Δ, the state-transition parameter A, the input/output projections B, C, or the recurrent state h itself – actually drives it?

We find H1 is the wrong frame for top-1 accuracy: AuM's full-precision accuracy already collapses to near-chance immediately past its training length (§4.1), leaving no headroom for quantization to measurably remove. AST, which degrades far more gradually, shows the only clear quantization-specific accuracy effect in the whole grid. For H2, a tensor-isolated ablation at native length (§4.2) finds two of AuM's four internal tensors genuinely fragile, with different failure shapes, while the other two are essentially quantization-proof.

2 Related Work

Models. Mamba [Gu and Dao, 2023] introduced input-dependent ("selective") SSM parameters as a linear-time alternative to attention. AuM [Erol et al., 2024] is the first purely SSM-based audio classifier, built as a bidirectional counterpart to AST [Gong et al., 2021]; both have official checkpoints fine-tuned on Speech Commands V2 [Warden, 2018], used directly here, with no retraining.

Quantization. General PTQ for large Transformers (e.g. GPTQ [Frantar et al., 2023]) targets Linear weights almost exclusively, since attention has no persistent numerical state to protect. Two recent works, MambaQuant [Xu et al., 2025] and Quamba2 [Chiang et al., 2025], build PTQ *recipes* specifically for the Mamba family and both independently flag the selective-scan's activations/state as the hard part to quantize – Quamba2 reports "massive outliers" in the scan's output activations and applies a dedicated per-state-group scheme to B, C to control them. We differ in kind, not just scale: rather than a deployable quantizer, we run a controlled ablation isolating *which single tensor* is responsible, as a function of input length, which these recipe papers do not study. Our finding that the state h is the most fragile tensor, with Δ fragile only past a bit-width threshold and B, C robust under plain per-tensor quantization, is consistent with Quamba2's choice to specially protect the scan's state/activations, and gives a tensor-level account of why.

3 Method

Models. AuM-Base (bidirectional, `bimamba_v1`, 24 layers, embed dim 768, AudioSet-pretrained then fine-tuned on Speech Commands V2) and AST (`MIT/ast-finetuned-speech-commands-v2`) – both official checkpoints for the identical 35-class task, avoiding a cross-dataset confound. 128×128-mel-bin log-spectrograms, 16×16 patches.

Data. Speech Commands V2 [Warden, 2018], 35 keyword classes. Native clips are $\sim$1s (both checkpoints' training length). Length-extended clips splice a native clip into the dataset's own official background-noise recordings at a random start/middle/end position, to 20/40/80/160s, keeping the noise characteristics dataset-native.

Standard quantization grid (H1). Static, symmetric fake quantize-dequantize simulation of every Linear layer's weights and/or activations: W8A16, W4A16 (weight-only), W8A8 (per-tensor, 3 seeds, and per-token). Calibration scales come from a held-out 256-clip split of the training set.

SSM-internal ablation (H2). AuM's fused selective-scan CUDA kernel cannot be hooked mid-recurrence, so we reimplement the scan step-by-step in PyTorch and validate it against the fused kernel first (max abs. diff. $\leq 6\times10^{-6}$ at native and 160s length). We then fake-quantize exactly one of $\{\Delta, A, B\&C, h\}$ to $\{8, 6, 4\}$-bit at a time, Linears at full precision throughout, isolating the SSM-internal effect from H1's. h is quantized at every recurrence step, per direction; Δ, A, B, C use one static per-layer scale from the same calibration set. Runs at native length only – the H1 grid shows every longer length is already at AuM's chance floor, so it carries no information here.

Metrics. Top-1 accuracy with a nonparametric bootstrap 95% CI (10,000 resamples over per-clip correctness). For H2, per-timestep state divergence $\|h_{q,t} - h_{fp,t}\|_2$, logged for every clip, layer, direction, timestep.

4 Results

4.1 H1: standard quantization vs. length

	native	20s	40s	80s	160s
AST full-prec.	1.000	0.594	0.362	0.178	0.041
AST W4A16	1.000	0.533	**0.270**	**0.098**	0.035
AuM full-prec.	0.981	0.089	0.073	0.067	0.051
AuM W4A16	0.971	0.079	0.083	0.064	0.064

Table 1: Top-1 acc., 3 clips/class (n=105 native, n=315/extended length). AST's W4A16 drop (bold) is real ($z\approx-2.5$ at 40s, $z\approx-2.9$ at 80s vs. full-prec.); AuM's quantized conditions stay within $\sim$1 SE of its own full-precision number everywhere – a confirmed null.

AuM's accuracy is already indistinguishable from chance ($1/35 \approx 2.9\%$) by 20s in full precision (Figure 1), leaving nothing for quantization to measurably remove. AST retains real accuracy to 80s and is the only model showing a reproducible, length-localized drop (Table 1): W4A16 is -9.2pp at 40s, -7.9pp at 80s, tracking *weight* bit-width specifically (every W8A8 variant shows no degradation beyond full precision). H1, read literally, is not supported by top-1 accuracy: AuM's problem is length generalization in full precision, not quantization.

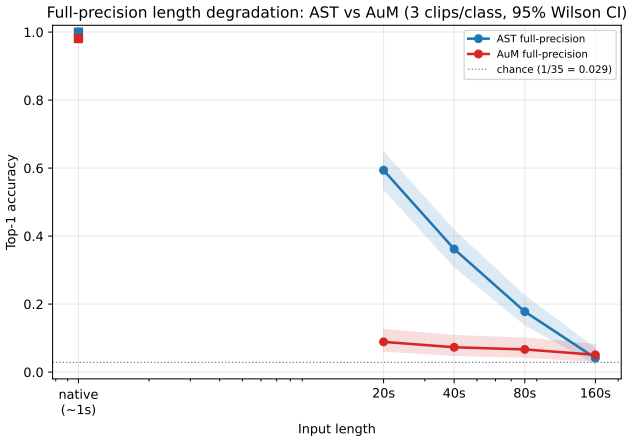


Figure 1: Full-precision accuracy vs. length. AuM collapses to near-chance immediately past training length; AST degrades gradually. Log-length slope, 20–160s: AST -0.612, AuM -0.040 – shallower only because AuM has already hit the floor.

4.2 H2: which SSM-internal tensor is fragile

Bits	Δ	A	B, C	state h
8	.957 [.921,.986]	.971 [.943,.993]	.964 [.929,.993]	**.836** [.771,.893]
6	.886 [.829,.936]	.971 [.943,.993]	.964 [.929,.993]	**.307** [.236,.386]
4	**.193** [.129,.257]	.950 [.914,.986]	.964 [.929,.993]	**.079** [.036,.129]

Table 2: AuM top-1 acc., native length, n=140 (35 classes, balanced), 95% bootstrap CI. A, B, C stay within noise of full precision throughout; h's 8-bit CI does not overlap A's/B, C's, so it is already significant at the mildest bit-width; Δ's 6- and 4-bit CIs do not overlap each other, confirming a real cliff.

A and B, C are robust at every bit-width tested – B, C's accuracy is identical (96.4%) across 8/6/4-bit. h and Δ are both fragile but shaped differently (Table 2, Figure 2): h degrades gradually, already significant at 8-bit; Δ is intact through 6-bit, then cliffs between 6- and 4-bit, landing near h's own 4-bit collapse (19.3% vs. 7.9%).

Per-timestep divergence $\|h_q - h_{fp}\|_2$ (n=436,800 values/condition) explains the shape difference. Under state quantization, median divergence rises smoothly (0.39 $\rightarrow$ 1.13 $\rightarrow$ 2.50 at 8/6/4-bit), consistent with error accumulating every one of $\sim$65 steps once h itself is corrupted. Under Δ quantization, the *median* at 4-bit (0.16) is actually *lower* than A's/B, C's own 4-bit medians (0.17, 0.26), but the *mean* and *max* are the largest of any condition at any bit-width (5.66, 126.8) – most steps barely perturbed, a minority blowing up.

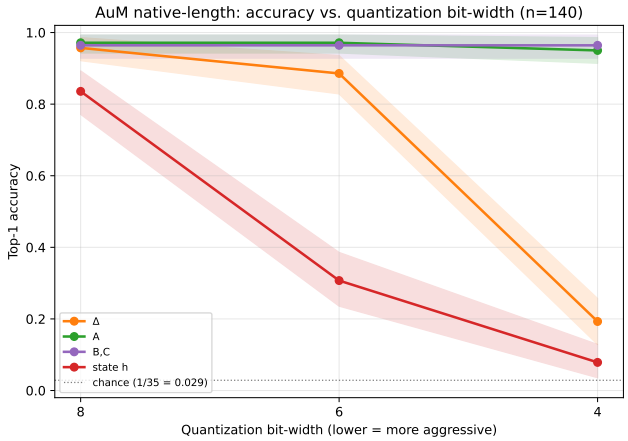


Figure 2: AuM native-length accuracy vs. SSM-internal bit-width. A, B, C nearly flat; h degrades from the mildest bit-width; Δ holds until a cliff between 6- and 4-bit.

This matches Δ's role gating $\bar{A}_t = \exp(\Delta_t A)$: a Δ error is usually harmless, but for some input-dependent values it pushes the recurrence into a regime where the error is exponentially amplified, and 4-bit pushes enough steps into that regime to break the clip.

5 Discussion and Limitations

This ablation uses one model pair, one dataset, and simple static per-tensor calibration – not the rotation- or clustering-based schemes MambaQuant/Quamba2 use to work around exactly this sensitivity. The 20–160s clips are native clips spliced into background noise, not naturally long speech, so they test out-of-distribution length generalization specifically. Divergence is an absolute L_2 norm, not normalized by $\|h_{fp}\|$, so cross-tensor magnitudes are only loosely comparable. The H2 ablation is native-length only by design, so we do not revisit whether this fragility compounds with length the way H1's grid does; a smaller, less class-balanced earlier run suggested divergence plateaus rather than compounds within a sequence, not re-verified at this sample size.

6 Conclusion

For audio SSM classifiers, "does quantization hurt the SSM more than the Transformer" is the wrong question once length pushes the SSM past its training distribution – it is already at chance in full precision. The right question is which part of the recurrence is load-bearing under quantization: the state h and the discretization gate Δ, not A or B, C. These two fail differently – h gradually, Δ as a bit-width cliff – a concrete target for future SSM quantization schemes, and a possible reason existing Mamba-specific PTQ recipes already single out the scan's state/activations for special treatment.

Reproducibility. All code, configs, calibration sets, per-clip raw results, and a full decision log (blockers, bugs, sample-size caveats included) are public.[2]

References

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[2] https://github.com/ayush11z/audio-ssm-quant